INCH-POUND

MS25455J
06 October 2020
SUPERSEDING
MS25455H
27 November 2003

PERFORMANCE SPECIFICATION SHEET

# RELAYS, ELECTROMAGNETIC, 5 AMPERES, 2 PDT, TYPE I, MAGNETIC LATCH, SOCKET MOUNTED, HERMETICALLY SEALED

INACTIVE FOR NEW DESIGN AFTER 5 JUN 1987.
NO SUPERSEDING SPECIFICATION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relay described herein shall consist of this specification and the latest issue of MIL-PRF-6106.

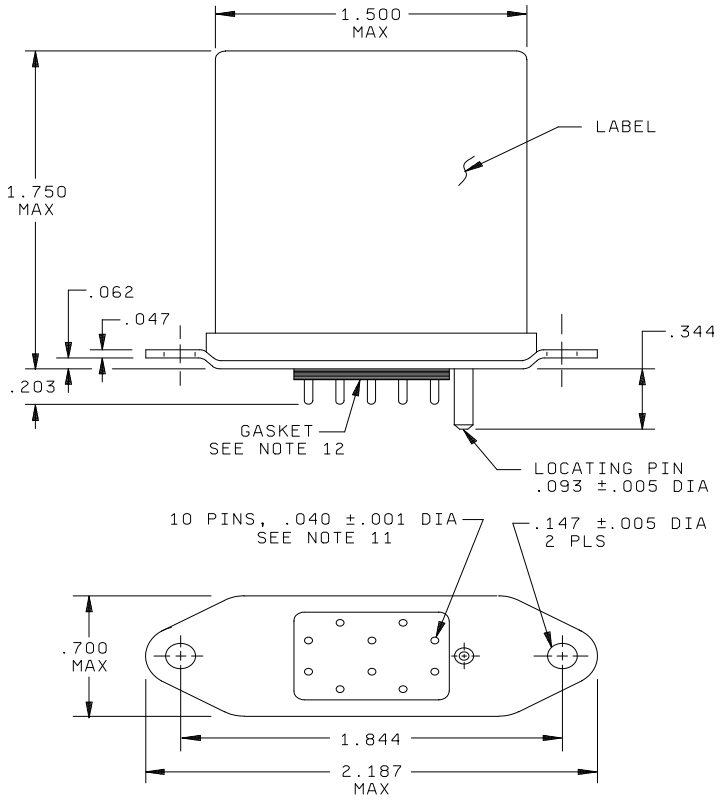
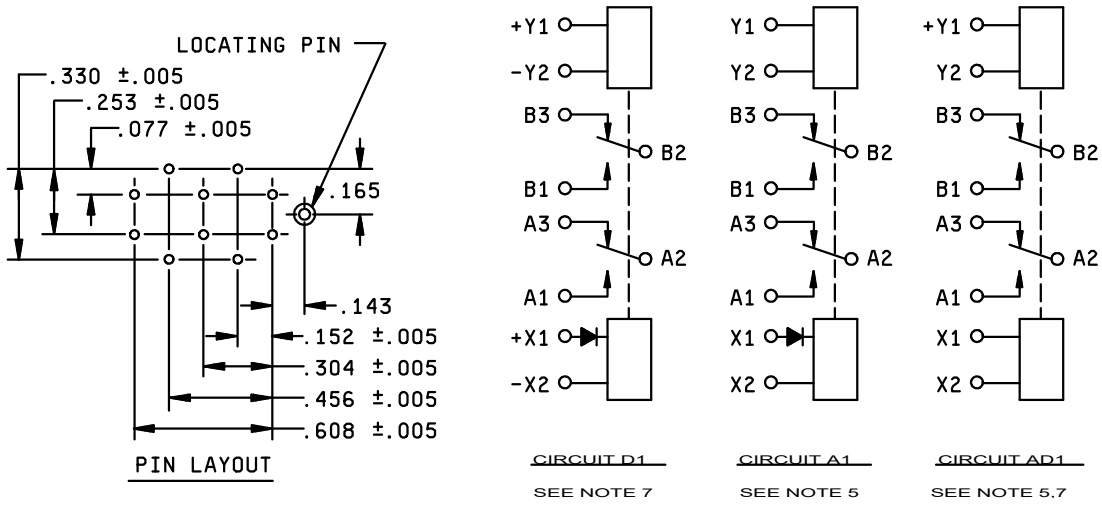



FIGURE 1. Dimensions and configurations.

AMSC N/A

FSC 5945

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| Inches | mm | Inches | mm |
|---|---|---|---|
| .001 | 0.03 | .165 | 4.19 |
| .005 | 0.13 | .203 | 5.16 |
| .010 | 0.25 | .253 | 6.43 |
| .040 | 1.02 | .304 | 7.72 |
| .047 | 1.19 | .330 | 8.38 |
| .062 | 1.57 | .344 | 8.74 |
| .075 | 1.91 | .456 | 11.58 |
| .077 | 1.96 | .608 | 15.44 |
| .093 | 2.36 | .700 | 17.78 |
| .143 | 3.63 | 1.500 | 38.10 |
| .147 | 3.73 | 1.750 | 44.45 |
| .152 | 3.86 | 1.844 | 46.84 |
| | | 2.187 | 55.55 |

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Terminal numbers need not appear on relay header provided there is affixed to the relay a suitable legible circuit diagram that permanently and positively identifies each terminal location specified hereon.
5. The use of diodes on ac relays is optional. Actual application must be shown on label.
6. Pins to be perpendicular to header surface within 1 degree.
7. Relay is magnetically latched in both positions. Caution note to observe polarity must appear on relays with dc coils.
8. Shock, vibration, and acceleration requirements applicable with coils de-energized.
9. In the event of conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.
10. Referenced Government documents of the issue listed in that issue of the Acquisition Streamlining and Standardization Information System (ASSIST) specified in the solicitation forms a part of this standard to the extent specified herein.
11. Socket pin terminals shall provide the operational, environmental, and interface characteristics to provide a reliable interconnect to gold-plated contacts. Terminals shall be gold plated. One system for gold plating that may be used is ASTM-B488, type 3, class 1.25 with a nickel underplate of 50 to 150 microinches thick. The gold plating system shall enable the product to meet the performance requirements of this specification and shall be approved by the qualifying activity.
12. Gasket shall provide a reliable seal between the relay and mating socket that will meet the environmental, operational, and interface requirements of the relay with the mating socket. The gasket shall have thickness .075, recessed .047 in to bracket. Gasket material according to SAE-AMS3332 has been considered acceptable.

FIGURE 1. Dimensions and configurations - Continued.



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TABLE I. Dash numbers and characteristics.

| Dash number MS25455- | Type | Coil | Terminal type | Max weight in pounds |
|---|---|---|---|---|
| D1 | I | dc | Plug in | 0.21 |
| A1 | I | ac | Plug in | 0.23 |
| AD1 | I | ac-dc | Plug in | 0.23 |

TABLE II. Operating characteristics.

| PIN MS25455- | Coil data | | | | | | | | | Time - milliseconds maximum | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Coil | Nominal | | | Max | | 1/ Max pick-up voltage | | | Op-erate 3/ | Re-lease 4/ | Contact Bounce | | | |
| | | | | | | | | | | | | Main | | Aux | |
| | | Volts 1/ | Freq Hz | Ω Res | Volts | Amp | Nor-mal 2/ | High temp test | Cont cur-rent test | | | NO | NC | NO | NC |
| D1 | X1, X2 Y1, Y2 | 28 | dc | N/A | 29 | 0.12 | 18 | 18 | 19.8 | 25 | N/A | 2 | 2 | N/A | N/A |
| A1 | X1, X2 Y1, Y2 | 115 | 400 5/ | N/A | 122 | 0.06 | 90 | 90 | 95 | 25 | N/A | 2 | 2 | N/A | N/A |
| AD1 | X1, X2 | 115 | 400 5/ | N/A | 122 | 0.06 | 90 | 90 | 95 | 25 | N/A | 2 | 2 | N/A | N/A |
| | Y1, Y2 | 28 | dc | N/A | 29 | 0.12 | 18 | 18 | 19.8 | 25 | N/A | 2 | 2 | N/A | N/A |

1/ CAUTION: Use of any coil voltage less than nominal coil voltage will compromise the operation of the relay.
2/ Over the temperature range.
3/ With nominal coil voltage.
4/ From nominal coil voltage.
5/ MS25455-A1 and -AD1 ac coils may be used on 60 Hz if maximum ambient temperature is limited to +85°C (coil current: 0.066 ampere maximum).

TABLE III. Rated contact load (amperes per pole) (case grounded).

| Type of load | Life operating cycles x $10^3$ | 28 V dc | | | | 115 V ac, 1 phase | | | | 115/200 V ac, 3 phase 1/ | | | | See appropriate notes |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Main | | Aux | | Main | | Aux | | Main | | Aux | | |
| | | NO | NC | NO | NC | 400 Hz | 60 Hz | 400 Hz | 60 Hz | 400 Hz | 60 Hz | 400 Hz | 60 Hz | |
| Resistive | 100 | 5 | 5 | | | 5 | 4 | | | | | | | |
| Inductive | 100 | | | | | | | | | | | | | |
| Inductive | 20 | 3 | 3 | | | 3 | 2 | | | | | | | |
| Motor | 100 | 1.5 | 1.5 | | | 1.5 | 1 | | | | | | | |
| Lamp | 100 | 0.8 | 0.8 | | | 0.8 | 0.6 | | | | | | | |
| Transfer load | | | | | | | | | | | | | | 2/ |
| Mechanical life reduced current | 400 | 1.25 | 1.25 | | | 1.25 | 1 | | | | | | | |
| Mixed loads | Applicable per specification MIL-PRF-6106 | | | | | | | | | | | | | |

1/ Absence of value indicates relay is not rated for 3-phase application.
2/ Transfer load indicates relay is suitable for transfer between unsynchronized ac power supplies at rating indicated.



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Environmental characteristics.

| | |
|---|---|
| Temperature range | -70°C to +125°C |
| Max altitude rating | 80,000 ft |
| Shock G-level | 50 g's |
| Duration | 11 ms |
| Max duration contact opening | 10 μs |
| Vibration - sinusoidal (see chart below) | |
| G-level | 10 g's |
| Frequency range | 5-1,500 Hz |
| Acceleration | 15 g's |

Electrical characteristics.

Insulation resistance:
Initial: 100 megohms.

After life or environmental tests: 50 megohms.

Dielectric strength (sea level).

| | Initial | After life tests |
|---|---|---|
| Coil to case | 1,000 V rms | 1,000 V rms |
| Aux contacts | | |
| All other points | 1,000 V rms | 1,000 V rms |

Dielectric strength (altitude).

| | (When mounted in mating socket) 80,000 ft |
|---|---|
| Coil to case | 500 V rms |
| Aux contacts | |
| All other points | 500 V rms |

Max contact voltage drop:
Initial: 0.150 volt.

After life test: 0.175 volt.

| | |
|---|---|
| Overload current | 20 amperes. |
| Rupture current | 25 amperes. |
| Duty rating | Continuous. |
| RFI specification | MIL-STD-461 |

(Applicable to coil circuits of ac operated relays).



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Conformance inspection.

Performance of groups B and C tests are not applicable.

Group A acceptance reports shall be submitted to the preparing activity on a yearly basis in order to retain qualification for this military specification sheet.

Qualification by similarity: See MIL-PRF-6106.

Referenced documents. In addition to MIL-PRF-6106, this document references the following:
MIL-STD-461 SAE-AMS3332 ASTM-B488

Custodians:
Navy - AS
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5945-2020-055)

Review activity:
Navy - EC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.



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